CS 532: Homework Assignment 2

Due: November 5th 6:00PM

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Collaboration Policy. Use of the Internet is allowed, but should not include searching for previous solutions or answers to the specific questions of the assignment. I will assume that, as participants in a graduate course, you will be taking the responsibility of making sure that you personally understand the solution to any work arising from collaboration.

Late Policy. 3% penalty for partial 24-hour period of delay.

Submission Format. Electronic submission on Canvas is mandatory. Submit in a zip file, a pdf file:

- source code (excluding libraries),
- points used in the computation,
- resulting images,
- at most one page of text explaining anything that is not obvious. Also include the code and output images/videos separately.

Problem 1: Teddy Stereo (40 points)

Download the Teddy stereo pair and ground truth from the course web page and implement a winner-take-all stereo algorithm using the census-transform (see Notes). The workflow is as follows:

For W_c in [3,7]
Compute the Census Transform images $\{I_{CT}\}$ using windows size of W_c x W_c.

Include census transform image in your report
For W_A in [5,9]

- **Compute disparity maps from $\{I_{CT}\}$ by aggregating per-pixel Hamming distances in neighborhoods of size W_A x W_A.**
- **Include disparity map image in your report**
- **Compute and report error rates**

Details:

- When showing in your report the census transform images, make sure to scale the range [min_Hamming_possible, max_Hamming_possible] to the range [0,255]. Pixels for which their census transform window (W_C) falls out of the image boundaries can be set to black. Discuss if the images look the same and why that is?
- Show the resulting disparity maps in the report. Pixels for which their aggregation window (W_A) falls out of the image boundaries can be set to black.
- The disparity range for the Teddy stereo pair is from 0 to 63. Read the ground truth disparity map and divide the values by 4 and round to the nearest integer. Compute the percentage of bad pixels (error rate) by counting the fraction of pixels that differs by more than one disparity level from the ground truth (divided by 4). Differences equal to 1 are considered acceptable. Report the error rates.

Debugging Hints:

- Using the same image as both the right and left image should give you a perfectly matching aggregated hamming distance for the case of zero disparity
- Running stereo with the second image being a horizontally shifted version of the first one (e.g padding the image with N zero-valued columns), should give you perfectly matching aggregated hamming distances at your chosen disparity=N

Problem 2: Fish Stereo (60 points)

Modify the image-rendering code provided for the first homework to render a pair of rectified images and compute the disparity map between them.

Step 1. Modify the Rendering code. You will need to understand how the example code works in more details and identify where and how to change it to achieve new capture examples. The required modifications to the existing code are:

a) Reduce the image pixel resolution to ¼ of the default size
b) Change the internal camera parameters so that the horizontal and vertical fields of view remain unchanged
c) Disable the part of the rendering code performing the 2D max_filter operation (used to "fill-in" blanks)

Step 2. Design the Stereo Geometry and Disparity Search Parameters. Control the pose of the camera (i.e. R and t) to generate/render a pair of rectified images from which to compute disparity. To do so you will need to analyze the scene geometry and camera positioning to make choices determining the following properties:

- Stereo camera baseline
- Disparity range to evaluate

Explain in your report how you justified your choices, explaining what are the minimum and maximum depths your disparity search is considering

Step 3. Run Stereo.

a) Convert the RGB image to greyscale using the following formula

Grey_value = (0.3)Red_Channel + (0.59) Green_Channel + (0.11)Blue_Channel.

b) Run the stereo method developed for problem 1 using a suitable range for disparities. Note the disparity range depends on the 3D scene and the camera baseline value. Show the resulting disparity maps in the report.